

8 Capabilities

8.1 Evaluation summary

Evaluation		Claude family models			Other models
		Claude Opus 5.5	Claude Opus 5	Claude Fable 5.1	GPT-6 Astra
SWE-bench Pro		89.9	79.2	81.2	–
SWE-bench Multilingual		93.9	89.5	89.1	–
SWE-bench Multimodal		61.4	59.4	54.7	–
FrontierCode v1.1 (Main)		54.4	48.0	50.3	53.3
Terminal-Bench 4.0		66.4	52.3	55.8	57.9
Terminal-Bench-Science 0.1		58.7	29.0	52.6	64.6
Humanity’s Last Exam	No tools	64.4	56.6	60.9	–
	With tools	67.7	63.6	65.6	57.2
OSWorld 2.0 (partial/strict)		81.8/48.7	74.0/37.2	80.7/42.8	–
HealthBench Professional		65.6	59.8	62.1	63.4
GDPval-AA v2.1		1846	1708	1735	1542
AA-Briefcase v1.1		1822	1673	1678	1569
AutomationBench		40.0	26.9	31.4	41.4

[Table 8.1.A] **Capability evaluation summary.** Unless otherwise noted, all Claude Opus 5.5 results use the following standard configuration: adaptive thinking at **max** effort, default sampling settings (temperature, top_p), averaged over five trials. Terminal-Bench 4.0 score is reported at **xhigh** effort. Context window sizes are evaluation dependent and do not exceed 1M tokens. The best score in each row is **bolded**. Competitor figures are drawn from the respective developers’ published system cards or benchmark leaderboards. See the [Claude Opus 5 System Card](#) for evaluation details of earlier Claude models.

8.2 SWE-bench Pro, Multilingual, and Multimodal

SWE-bench tests AI models on real-world software engineering tasks. We report three variants, where each score is an average over five trials:

- **SWE-bench Pro**¹⁶ is composed of problems drawn from actively maintained repositories with large, multi-file diffs. Claude Opus 5.5 achieved 89.9%.
- **SWE-bench Multilingual** extends the format to 300 problems across nine programming languages. Opus 5.5 achieved 93.9%.
- **SWE-bench Multimodal**¹⁷ adds visual context (screenshots and design mockups) to the issue descriptions. Opus 5.5 achieved 61.4%.

8.3 DeepSWE v1.1

DeepSWE is a set of 113 long-horizon software engineering tasks built to measure the capabilities of frontier coding agents. The tasks are diverse, reflect real-world complexity, and are written from scratch to avoid benchmark contamination. Claude Opus 5.5 scored an average of 74.2% over five trials.

8.4 FrontierCode

FrontierCode is an agentic coding benchmark of 150 software engineering tasks created by Cognition. Tasks are derived from real pull requests in open-source repositories, such as fixing websocket bugs in `aiohttp`, hardening Prisma's browser bundle, or extending JSON schema linting rules. Each task gives the agent a checked-out repository, an issue description, and the repository's testing, lint, and style guidelines; the agent then works autonomously in a containerized environment with internet access to produce a final patch, with no human intervention and no timeout information. Patches are graded against blocking functional criteria (primarily held-out unit tests, plus scope and performance blockers) plus weighted rubric criteria, including model-graded checks for required test coverage and prohibited implementation patterns. Tasks were authored by maintainers of the underlying repositories and individually reviewed by Cognition researchers, with a random subset manually solved to verify fairness. We report FrontierCode's overall score, a composite measure that grades each patch on blocking functional criteria (held-out unit tests) together with weighted code-quality rubric criteria, as mean@5.

¹⁶ Deng, X., et al. (2025). SWE-Bench Pro: Can AI agents solve long-horizon software engineering tasks? arXiv:2509.16941. <https://arxiv.org/abs/2509.16941>

¹⁷ Yang, J., et al. (2024). SWE-Bench Multimodal: Do AI Systems generalize to visual software domains? arXiv:2410.03859. <https://arxiv.org/abs/2410.03859>

and container communication protocol), ultimately making Terminal-Bench a more stable and rigorous evaluation.

Claude Opus 5.5 scored 66.36% on Terminal-Bench 4.0 with safeguards enabled; requests flagged by the safeguards were answered by a fallback model following the default server-side fallback policy (2.5% of requests, affecting 10% of trials). For comparison, Claude Mythos 5.1 scored 60.9% and Claude Fable 5.1 scored 55.8%, whereas Claude Opus 5 scored 52.3%. Scores are averaged over five trials per task (330 trials) for Opus 5.5, 10 trials per task (660 trials) for Claude Mythos 5.1, and 15 trials per task (990 trials) for Opus 5. Treating trials as independent, the standard error is ± 2.6 points for Opus 5.5 and ± 1.6 -2 points for the other models. The numbers reported use Claude Code in `--bare` mode and `xhigh` thinking effort for Opus 5.5 (which at `max` effort scores 64.8%, within noise of `xhigh`), and `max` effort for the comparison models.

For reference, the public leaderboard reports Claude Opus 5 at 51.8% (five trials, using the Claude Code harness), within the noise of our internal setup. OpenAI's GPT-6 Astra reported their final Terminal-Bench 4.0 score at 57.9%, scored at `high` thinking effort (in their reporting `max` effort fared slightly lower than high, so we picked their highest number).

8.6 Terminal-Bench-Science 0.1

Terminal-Bench-Science is a Stanford-led community benchmark of 70 tasks drawn from scientific research workflows, authored and reviewed by scientists and researchers across the life, physical, earth, mathematical, and engineering sciences. Scientific advisors to the project are from MIT, Princeton, University of Washington, Genentech, and Stanford.

Tasks are set up as typical agentic problems: an agent CLI works in a self-contained environment from a natural language instruction and is graded against hidden, task-specific tests on the artifacts it produces.

Claude Opus 5.5 scored 58.7% on Terminal-Bench-Science with safeguards enabled; requests flagged by the safeguards were answered by a fallback model following the default server-side fallback policy (3.9% of requests, affecting 5% of trials). It averaged over 10 trials per task (700 trials). Claude Fable 5.1 scored 52.6% (10 trials per task, 700 trials), Claude Opus 5 scored 29.0% (12 trials per task, 840 trials), and Claude Fable 5 scored 24.7% (10 trials per task, 700 trials). The standard error, clustered by task, is ± 3.5 -4.8 points (± 4.8 for Opus 5.5).